NSF I-Corps
Market Sizing Analysis Worksheet

NSF I-Corps Team:	**003 Fruitful Network Development**				
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Market Sizing Analysis – *Type your answers in the gray boxes provided below – market value definitions and sample analysis are available toward the bottom of this document.*

It is important to understand the overall market opportunity for your product, service, or technology. Potential partners, investors, and other stakeholders need to know that the overall market opportunity is large enough to warrant their attention and that you understand your product's specific portion of that opportunity.

Market sizing analysis requires research and assumptions tied to your ongoing customer interviews. It may simplify the process if your think in terms of *Yearly Sales and populations in the United States Market*

It is not necessary to pay for market research at this point in the process. Use an initial Google search to learn the big picture. Local and university libraries may provide access to market data for this exercise.

Other Resources:

- Steve Blank's video, *How to Build a Startup, Total Available Market* https://youtu.be/_-N5u7i_Vgk
- Detailed information. including definitions, descriptions, and a sample analysis, can be found at the end of this worksheet.

Part 1 - The Big Picture

Type your answer in the boxes provided.

What is your Total Available Market? (TAM) in USD$	≈ $30–$70 Billion Dollars
Sources and Rationale	By considering how much greater the agricultural output of industrial farms is versus family owned with respect to the amount of land used to produce products with short shelf lives or that require refrigerated transport chains. (Compare for in the same agricultural product type to avoid an answer that omits a bias for an output comparison that may be affected by family owned farms having a tendency to produce different products, in order to focus on differences in output that can assumed to be attributed to factors such as waste, distribution, economies of scale, ect.) - According to USDA Economic Research Service figures on lettuce producers, large-scale industrial operations (over 1,000 acres) net about 20% higher returns per acre (e.g. around $3,000 vs. $2,400) than smaller family farms (under 100 acres). This margin primarily reflects more efficient cold-chain logistics, distribution networks, and economies of scale that reduce spoilage and lower per-unit handling costs (USDA ERS 2021). - Given that 1,843,000/1,900,000 (97%) of farms and 919,784.375/1,372,812.5mi^2 (67%) of farmland is owned by individuals and Family farms with an annual gross cash farm income (GCFI) of at least a million dollars account for $260.688/543.1B (48%) of the value of farm production. - The agricultural output revenue per acre of a family farm varies widely depending on factors like crop type, farm size, and location, but small family farms, which account for 86% of US farms, operate on 41% of agricultural land and produce 17% of the total value of production Therefore, if family-owned operations matched the 15–25% per-acre value advantage seen in industrial-scale lettuce production (largely from better cold-chain logistics and distribution), a rough extrapolation suggests an additional $30–$70 billion in overall annual production value could be realized across comparable short-shelf-life crops in aggregate. This broad range reflects potential efficiency gains in areas such as handling costs, transportation, and reduced spoilage.

How many customers (people, businesses, organizations) are in the TAM?	≈ 1,843,000
Sources and Rationale	(See above)

How often will they buy?	Initial Cost (1) + Maintenance and updating (≈ 8/year) + (Maybe)Hosting(12/year)
Sources and Rationale	By product and model nature; certainly initial web development, confidently assumed maintenance and updating, and moderate confidence in assuming optional hosting and MSP.

How much would they pay for your product?	≈ 300$(x1) + 20$(12/year)
Sources and Rationale	Weeks Minimum Wage + (MSP COGS)(Shopify, Wix, Square Space, Go daddy approx. rates) (Maybe for a safer estimate it should be assumed based on that a healthy net profit margin for an MSP company typically ranges from 20% to 30%, with some best-in-class firms aiming for 35% or higher, while a gross profit margin of 50-60% is considered industry average.)

Part 2 - Your Customer Segment

Type your answer in the boxes provided.

What is your Serviceable Available Market (SAM) in USD$	≈ $1.0–$2.5 Billion Dollars
Sources and Rationale	[($16,277.81/farm) - ($37,981.55/farm)] * (66,217) = $1,077,867,607.20 - $2,515,024,416.70 (Ratio from sources and rational above and SAM below)

How many customers are in your SAM?	≈ 66,217
Sources and Rationale	According to the USDA Ag. Stat. Service: In 2022, Ohio had 66,217 family farms, representing about 87% of all farms in the state, with the remaining farms being partnerships, corporations, or other entities.

Part 3 - Your Initial Target

Type your answer in the boxes provided.

What is your Serviceable Obtainable Market (SOM) in USD$	≈ 3.1 million for 99.1565% of the market with in 4 years
Sources and Rationale	If I assume an natural growth rate for serviceable parties X: Let X = SUM(Summit, Geauga, Cuyahoa, Portage) = 2,727 farms X / 52 (*/number of week in a year assuming one service a week*/) ≈ 52.44 years ln(52.44) ≈ 3.95 */years to total initial servicing*/ Therefore, in about 4 years revenue to me made for initial service can be found by: 4 * 19 (*/Highest hosting price still lower than competitors*/) * 12 (*/months in a year*/) = 2.5 million dollars in S_initial chargers So assuming a book of accounts growth of for S_i: Σ[e^(nt)] = $621,756 | for t ∈ [0, 3.9597 years] Then over the course of of approximately 4 years there is available market of $2,487,024 + S_i ≈ $3,108,790

How many customers are in your SOM?	≈ 2,727
Sources and Rationale	County farms ({Summit, Geauga, Cuyahoa, Portage}): 392 + 1,133 + 104 + 1,118 = 2,727 2,727

Part 4 – Top-Down Analysis

Type your answer in the boxes provided.

Total Available Market *(from Part 1)*	≈ $30–$70 Billion
Serviceable Available Market *(from Part 2)*	≈ $1.0–$2.5 Billion
Serviceable Obtainable Market *(from Part 3)*	≈ $ 3.1 Million

Part 5 – Bottom-Up Analysis

Type your answer in the boxes provided

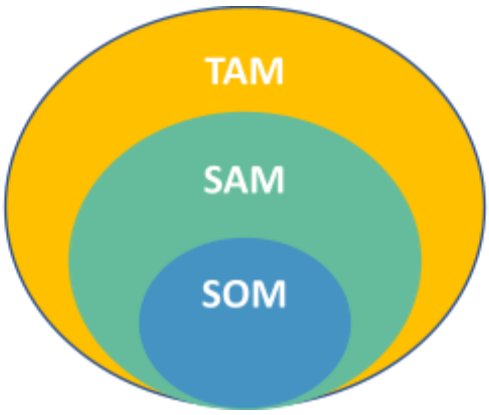
Price per unit	≈ 300$(x1) + 20$(12/year)
Sources and Rationale	(See above)

Number of target customers	≈ 2727
Sources and Rationale	(See above)

How often they buy	Initial Cost (1) + Maintenance and updating (≈ 8/year) + (Maybe)Hosting(12/year)
Sources and Rationale	(See above)

Annual Revenue from target market *(Multiply price per unit X number of target customers X how often they buy)*	first year ≈ > = 22,256 (customer per week for S_initial = $10,400 & initial years fees for hosting will be >= $11,856)

What is TAM, SAM, SOM?



TAM - overall revenue opportunity that is available to a product or service if 100% market share was achieved.

SAM – portion or segment of the overall market that a business can acquire based on its business model.

SOM – estimated portion of the serviceable market that an organization predicts it will capture at start of business.

TAM (Total Available Market *sometimes referred to as total Attainable or Addressable market.)*
The overall revenue opportunity that is available to a product or service if 100% market share was achieved.

- TAM helps determine the level of effort and funding that should put into a new idea or business line.
- TAM offers high-level insight on growth, funding, and investment potential.

SAM (Serviceable Available Market *sometimes referred to as Served Attainable or Addressable market.)*
The portion or segment of TAM that a business can acquire based on its business model.

- A business is unlikely to address the total market due to limitations in business model such as specialization or geography.
- SAM is most useful for businesses to objectively estimate the portion of the market they can use to determine their target market.

SOM (Serviceable Obtainable Market *sometimes referred to as target market or share of the market.)*
Represents the estimated portion of revenue an organization predicts it can capture within a specific product segment.

- It is crucial to measure the serviceable obtainable market to determine how many customers would ***realistically benefit*** from buying your company's product or service.
- SOM is most useful to determine short-term growth targets.

As you move forward with your commercialization plan you may need to conduct a **Top-Down or Bottom-up Market Analysis**. Once you've found your TAM, SAM, and SOM, these analyses are easy to produce.

Top-Down Analysis starts with the overall size of a market that includes your product, service, or technology. You will then break that overall market down to estimate your specific market more closely. In our coffee shop example on the next page, we started with the entire coffee market in the US. A smaller percent of coffee market is actually purchased or consumed at a coffee shop. Coffee shops are also hyper-local, so we further filtered sales to a specific college campus. Top-Down numbers come directly from your TAM, SAM, and SOM research.

Bottom-Up Analysis works in the opposite direction. Bottom-up starts with the smallest units of sales (# of customer, # of units, average price, etc.) When you perform a "Bottom-Up" analysis, you'll use those small units or building blocks to calculate market size – in the coffee shop sample below, we start with price per cup of coffee. Multiply by the number of customers in the geographic region and then multiply by the number of times they'll buy coffee from your shop.

Best Practice Tip: Initial market sizing is more of an art than a science. Keep track of the data sources, assumptions, and rationale behind your market sizing so you can tell the story of your market and easily update numbers as you conduct more discovery and learn more about your market.

Other Resources:

- Steve Blank's video, *How to Build a Startup, Total Available Market* https://youtu.be/_-N5u7i_Vgk
- Campus Coffee Shop Example below

Market Sizing Example
Opening an On-Campus Coffee Shop (*Calculations are approximate*)

Part 1 - The Big Picture

Type your answer in the boxes provided.

What is your Total Available Market? (TAM) in USD$	**$80,000,000,000**
Sources and Rationale	Used Statista Database to determine total US coffee market as well as the percent that comes from coffee shops and % of the population that drink coffee US coffee shop industry market value 2022 | Statista

How many customers (people, businesses, organizations) are in the TAM?	**333,000,000**
Sources and Rationale	Population data from Census Bureau Data.

How often will they buy?	**365**
Sources and Rationale	One time per calendar day

How much would they pay for your product?	**$3.00 cup / $77 per year**
Sources and Rationale	Researched prices at 5 local coffee shops to find average price per cup. Found the average spend per person/per year on Statista

Part 2 - Your Customer Segment

Type your answer in the boxes provided.

What is your Serviceable Available Market (SAM) in USD$	**$25,000,000,000**
Sources and Rationale	US coffee shop industry market value 2022 | Statista (percent of coffee sales from coffee shops – most coffee is purchased for home or institutional/business consumption)

How many customers are in your SAM?	**130,400,000**
Sources and Rationale	US coffee shop industry market value 2022 | Statista (40% percent of population that drinks coffee)

Part 3 - Your Initial Target (SOM)

Type your answer in the boxes provided.

What is your Serviceable Obtainable Market (SOM) in USD$	**$1,200,000** (Rounded to nearest 100K)
Sources and Rationale	Share of market based on population on campus

How many customers are in your SOM?	**15,000 students and staff on campus.**
Sources and Rationale	Checked University website and verified through admin office

How often do they buy per year	**205**
Sources and Rationale	Based on academic calendar of "active" class days.

Part 4 – Top-Down Analysis

Type your answer in the boxes provided.

Total Available Market *(from Part 1 above)*	**$80B**
Serviceable Available Market *(from Part 2 above)*	**$25B**
Serviceable Obtainable Market *(from Part 3 above)*	**$1.2M** (rounded to nearest 100, 000)

Part 5 – Bottom-Up Analysis

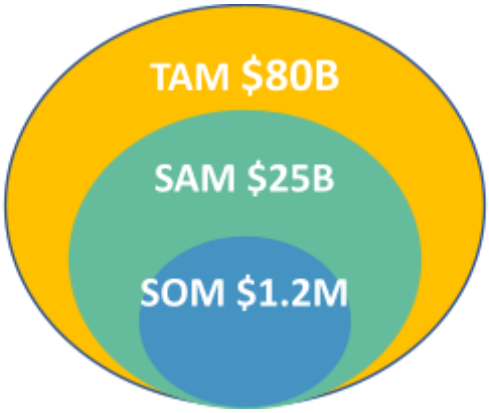
Type your answer in the boxes provided.

Price per unit	**$3.00**
Sources and Rationale	Checked online, local coffee shops, and other college campuses

Number of target customers	**1875**
Sources and Rationale	Based on % of campus population (students and staff) that buy coffee from coffee shops

How often they buy	**205**
Sources and Rationale	Based on academic calendar of "active" class days. Assumes 1 purchase per day

Annual Revenue from target market	**$1,153,125**
Sources and Rationale	$3.00 x 1875 x 205 = 1,153,125 *(Multiply price per unit X number of target customers X how often they buy)*



Sample Campus Coffee Shop Data Table

You may find it helpful to track your data in an excel workbook. This makes it easy to make quick changes as you learn more about your market. Feel free to track data in a manner that works best for you.

	A	B	C	D	E
1	**EXAMPLE: Coffee House Market Sizing**				
2		TAM	SAM	SOM	
3		Total Addressable Market	Servicable Attainable Market	Servicable Obtainable Market	Source/Reasoning
4					
5	Us Coffee Market	$80,000,000,000			Statista
6	US Population	326000000			US Census
7	Coffee Spend Per Capita	$245			*Coffee Market divided by Population*
8	Coffee House Market		$25,000,000,000		Statista
9	Coffee House Spend Per Capita		$77		*Coffee House Market divided by Population*
10	Students/Staff			15000	Administrative Office
11	Campus Coffee House Spend			$1,150,307	*Coffee House Spend per Capita multiplied by Students/Staff*
12					
13	**Top Down Market Analysis**				
14	US Market	$80,000,000,000	TAM		*Top Down and Bottom Up Analysis*
15	Coffee Shop Market	$25,000,000,000	SAM		*don't need to be exactly the same,*
16	UAkron Coffee Shop	$1,150,307	SOM		*but should be similar*
17					
18	**Bottom Up Market Analysis**		Population	$$	Source/Reasoning
19	Avg per Cup	$3.00		$3.00	*Average of local prices*
20	Number of Students/Staff	15000			Campus Website
21	% that drink coffee	40%	6000		Statista
22	% that buy from Coffee House	31%	1875		Statista (Verified through interviews)
23	Revenue per day	$5,625		$5,625	Calculation
24	Number of days that staff and students are actually on campus	205			Administrative Office
25	Revenue per year	$1,153,125		$1,153,125	
26					